## C4.6 Clean transactions

This section defines the state changes associated with the Clean transaction group that are issued on the AR channel. The Clean transactions are:

- *CleanUnique*
- *CleanShared* on page C4-198
- *CleanInvalid* on page C4-199.

### C4.6.1 CleanUnique

A CleanUnique transaction is used in a region of memory that is shareable with other masters. The CleanUnique transaction ensures that:

- The cache line can be held in a Unique state. This permits the master to carry out a store operation to the cache line, but the transaction does not obtain a copy of the data for the master.
- Data held in another cache in a Dirty state is written to main memory and all other copies of the cache line are removed.

Typically, a CleanUnique transaction is used before a partial cache line store operation to shareable memory when the master already has a copy of the data.

The transaction response requirements are:

- the IsShared response must be deasserted to indicate that the cache line is unique
- the PassDirty response must be deasserted.

**Note**

The cache line state changes associated with the CleanUnique transaction that Table C4-13 and Table C4-14 on page C4-198 show, do not include the cache line state changes associated with any subsequent store operation by the master once the cache line is in a Unique state.

Table C4-13 shows the expected cache line state changes for the CleanUnique transaction:

**Table C4-13 Expected CleanUnique cache line state changes**

| Transaction | Start state | RRESP[3:2] IsShared/PassDirty | Expected end state | Legal end state | |
|---|---|---|---|---|---|
| | | | | With Snoop Filter | No Snoop Filter |
| CleanUnique | SC | 00 | UC | UC, SC | I, UC, SC |
| | SD | 00 | UD | UD, SD | UD, SD |


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Table C4-14 shows other permitted cache line state changes for the CleanUnique transaction:

**Table C4-14 Other permitted CleanUnique cache line state changes**

| Transaction | Start state | RRESP[3:2] IsShared/PassDirty | Expected end state | Legal end state With Snoop Filter | Legal end state No Snoop Filter |
|---|---|---|---|---|---|
| CleanUnique | I | 00 | I | I[a] | I |
| | UC | 00 | UC | UC, SC | I, UC, SC |
| | UD | 00 | UD | UD, SD | UD, SD |

a. See *Snoop filter cache line allocation awareness*.

On completing a CleanUnique transaction, the initiating master has permission to store to the cache line. If the cache line was in the Invalid state before the store operation, then the store must be a full cache line size for the cache line to be allocated in the cache. After the full cache line store, the cache line is in the UniqueDirty state. The store must occur atomically with the completion of the CleanUnique transaction. Therefore, any snoop that occurs after the CleanUnique transaction must be delayed until the store is complete.

CleanUnique transactions can be used for Exclusive accesses, see Chapter C9 *Exclusive Accesses*.

### Snoop filter cache line allocation awareness

A snoop filter regards a cache line as allocated after the completion of a CleanUnique transaction. Therefore, the snoop filter has the correct information on the allocation of a cache line in the following circumstances:

- The cache line was allocated before the CleanUnique transaction and remains allocated after the CleanUnique transaction completes.
- If the cache line was not allocated before the CleanUnique transaction, or the cache line was invalidated during the CleanUnique transaction, then when the CleanUnique transaction completes the master:
  - Performs a full cache line store and the cache line is allocated.
  - Performs a WriteBack transaction of either a full or partial cache line store, and indicates to the snoop filter that the cache line is no longer allocated.
  - Reissues another transaction, for example a ReadUnique transaction, before performing a full or partial cache line store and the cache line becomes allocated.
  - Does not perform a store operation. In this situation the master must issue an Evict transaction to indicate to the snoop filter that the cache line is no longer allocated.

## C4.6.2 CleanShared

A CleanShared transaction is a broadcast cache clean operation. It can be used in shareable and non-shareable memory regions.

A CleanShared transaction is used to ensure all cached copies of a main memory location are clean.

**Note**

If the master carrying out the cache maintenance operation holds the cache line in a Dirty state then the master must carry out a WriteBack or WriteClean transaction so that the cache line is in a Clean state before it issues a CleanShared transaction.

The transaction response requirements are:

- the IsShared response indicates if the cache line is shared or unique
- the PassDirty response must be deasserted.


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Table C4-15 shows the expected cache line state changes for the CleanShared transaction:

**Table C4-15 Expected CleanShared cache line state changes**

| Transaction | Start state | RRESP[3:2] IsShared/PassDirty | Expected end state | Legal end state With Snoop Filter | Legal end state No Snoop Filter |
|---|---|---|---|---|---|
| CleanShared | I | 00 | I | I | I |
| | | 10 | I | I | I |
| | UC | 00 | UC | UC, SC | I, UC, SC |
| | SC | 00 | UC | UC, SC | I, UC, SC |
| | | 10 | SC | SC | I, SC |

### C4.6.3 CleanInvalid

A CleanInvalid transaction is a broadcast cache clean and invalidate operation. It can be used in shareable and non-shareable memory regions.

A CleanInvalid transaction is used to ensure that main memory is updated and there are no cached copies of a main memory location.

**Note**

If the master carrying out the cache maintenance operation holds the cache line in a Dirty state, then the master must carry out a WriteBack or WriteClean transaction, and then invalidate the cache line, so that the cache line is in the Invalid state before it issues a CleanInvalid transaction.

The transaction response requirements are:

- the IsShared response must be deasserted
- the PassDirty response must be deasserted.

Table C4-16 shows the expected cache line state changes for the CleanShared transaction:

**Table C4-16 Expected CleanInvalid cache line state changes**

| Transaction | Start state | RRESP[3:2] IsShared/PassDirty | Expected end state | Legal end state With Snoop Filter | Legal end state No Snoop Filter |
|---|---|---|---|---|---|
| CleanInvalid | I | 00 | I | I | I |


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